

	Case Name: Railway Station Antwerp	Sector	Construction (civil)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
Submitted by	Matilde Casado and Margarida Antunes	Risk Analysis	Schedule with costs	
Date	2015		Random simulation	
File Name	C2015-27 Railway Station Antwerp		One of nine std. scenarios	
		Project Control	User defined distributions	
			Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

A renovation project for Antwerp Railway Station, including painting, repairs, and upgrades in multiple areas.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	18	Serial/Parallel (SP)	23%
Planned Duration (PD)	68 days*	Activity Distribution (AD)	30%
Budget At Completion (BAC)	22 703.5 €	Length of Arcs (LA)	0%
Renewable Resources	11	Topological Float (TF)	52%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	20.7	12.0	-0.3	CI	27.7	43.3	-1.1
CRI-rho	19.1	12.0	-0.7	SI	42.3	40.2	-0.6
CRI-tau	12.2	7.8	-0.7	SSI	12.7	22.6	-2.0
				CRI-r	15.9	17.1	-2.0
				CRI-rho	16.0	17.1	-2.0
				CRI-tau	11.4	11.9	-1.9

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	29.3	-28.6	1	2.9	0.8
PV - SPI	21.3	18.6	CPI	4.1	1.8
PV - SCI	23.4	19.2	SPI	22.7	22.7
ED - 1	38.2	-37.1	SPI(t)	21.7	21.7
ED - SPI	21.3	18.6	SCI	22.9	22.9
ED - SCI	21.8	18.9	SCI(t)	21.9	21.9
ES - 1	35.0	-34.6	0.8 CPI + 0.2 SPI	11.7	11.6
ES - SPI(t)	19.2	18.1	0.8 CPI + 0.2 SPI(t)	10.5	10.4
ES - SCI(t)	20.0	18.4			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 6 tracking periods with a length of approximately 2 weeks in the first 3 tracking periods and 1 month in the last ones. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-3153.4	-1277.7	-71.8	0.8	0.9	0.8	1.0
std dev	1123.6	1177.3	56.8	0.1	0.1	0.2	0.0
final	-2609.6	0.0	-112.0	0.9	1.0	0.8	1.0

2.3.3.2. Time forecasting

PD	68 days	Real Duration	82 days	Late	20.6%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	73.3	1.6	1.8	1.8
PV - SPI	74.0	1.3	1.6	1.6
PV - SCI	74.7	1.0	1.5	1.5
ED - 1	75.3	1.0	1.4	1.4
ED - SPI	76.0	1.3	1.2	1.2
ED - SCI	76.7	1.6	1.1	1.1
ES - 1	77.5	1.9	0.9	0.9
ES - SPI(t)	78.3	1.6	0.7	0.7
ES - SCI(t)	79.0	1.3	0.6	0.6

2.3.3.3. Cost forecasting

BAC	22 703.5 €	Real Cost	25 313.1€	Over Budget	10.3%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	25856.9	1123.6	0.4	-0.4
CPI	30684.8	5525.7	3.5	-3.5
SPI	28472.5	3522.7	2.1	-2.1
SPI(t)	30070.5	6199.2	3.1	-3.1
SCI	34798.9	10217.2	6.2	-6.2
SCI(t)	37430.1	14276.8	8.0	-8.0
0.8 CPI + 0.2 SPI	30177.1	5047.1	3.2	-3.2
0.8 CPI + 0.2 SPI(t)	30527.3	5586.1	3.4	-3.4